Topic 1 Review Set 2

Module 3 Topic 1 · weekend practice on Wayground · 20 multiple choice

1. Carlos has $25 in savings and adds $18 each week. Which equation gives the amount, y, he has after x weeks? · Write an equation

A. $y = 25x + 18$

B. $y = 18x + 25$

C. $y = 18x - 25$

D. $y = 43x$

2. A phone card starts with $50. Each call costs $2.50. Which equation gives the money left, y, after x calls? · Write an equation

A. $y = 2.5x + 50$

B. $y = 50x - 2.5$

C. $y = -50x + 2.5$

D. $y = -2.5x + 50$

3. Ana buys 3 adult tickets at $9.50 each and some student tickets at $6.25 each. Which equation gives the total cost, y, for x student tickets? · Write an equation

A. $y = 6.25x + 28.50$

B. $y = 9.50x + 6.25$

C. $y = 28.50x + 6.25$

D. $y = 15.75x$

4. Solve: $4(x - 3) = 2x + 6$ · Solve

A. $x = 3$

B. $x = -3$

C. $x = 9$

D. $x = 1.5$

5. Solve: $0.5(6x + 8) = 2x - 1$ · Solve

A. $x = 5$

B. $x = -3$

C. $x = -5$

D. $x = 3$

6. What value of x makes $7 - 3(x - 2) = 2x - 2$ true? · Solve

A. $x = 3$

B. $x = -3$

C. $x = 1$

D. $x = 5$

7. How many solutions does $2(3x + 4) = 6x + 8$ have? · How many solutions

A. No solution

B. One solution, $x = 0$

C. One solution, $x = 8$

D. Infinitely many

8. Which equation has NO solution? · How many solutions

A. $5(x + 1) = 5x + 5$

B. $5(x + 1) = 5x - 2$

C. $5(x + 1) = 4x - 2$

D. $5x + 1 = 4x + 1$

9. Solve: $-3x + 5 > 17$ · Inequality

A. $x < -4$

B. $x > -4$

C. $x < 4$

D. $x > 4$

10. Solve: $2(5 - x) \leq x - 8$ · Inequality

A. $x \leq 6$

B. $x \geq -6$

C. $x \geq 6$

D. $x \leq -6$

11. Which describes the graph of $n > -2$? · Graph an inequality

A. Closed circle at -2, shade right

B. Open circle at -2, shade right

C. Open circle at -2, shade left

D. Closed circle at -2, shade left

12. Which describes the graph of $x \leq 4$? · Graph an inequality

A. Open circle at 4, shade left

B. Closed circle at 4, shade right

C. Open circle at 4, shade right

D. Closed circle at 4, shade left

13. A club already has $60 and earns $40 per car wash, w. They need at least $500. Which inequality fits? · Write an inequality

A. $40w + 60 \geq 500$

B. $40w + 60 \leq 500$

C. $60w + 40 \geq 500$

D. $40w + 60 > 500$

14. Using $40w + 60 \geq 500$, how many car washes does the club need? · Write an inequality

A. At most 11

B. At least 11

C. At least 12

D. At least 14

15. Rewrite in slope-intercept form: $3x + 4y = 12$ · Slope-intercept form

A. $y = 3/4\ x + 3$

B. $y = -3x + 12$

C. $y = -3/4\ x + 3$

D. $y = -4/3\ x + 4$

16. Rewrite in slope-intercept form: $6x - 2y = 10$ · Slope-intercept form

A. $y = -3x + 5$

B. $y = 3x + 5$

C. $y = 6x - 10$

D. $y = 3x - 5$

17. Rewrite in slope-intercept form: $y + 2 = 4(x - 1)$ · Slope-intercept form

A. $y = 4x - 2$

B. $y = 4x + 2$

C. $y = 4x - 3$

D. $y = 4x - 6$

18. Solve $y = mx + b$ for x. · Literal equation

A. $x = (y - b)/m$

B. $x = y/m - b$

C. $x = (y + b)/m$

D. $x = m(y - b)$

19. Solve $3x + 2y = z$ for y. · Literal equation

A. $y = z/2 - 3x$

B. $y = (z - 3x)/2$

C. $y = (z + 3x)/2$

D. $y = 2(z - 3x)$

20. A square has sides of $(x + 2)$ cm. A rectangle is $(x + 6)$ cm long and 2 cm wide. Their perimeters are equal. What is x? · Geometry

A. $x = 2$

B. $x = 8$

C. $x = 4$

D. $x = 6$